

# Killara High School



## Extension 1 Mathematics Term 1 2009

### General Instructions

- Working Time – 2 hours
- Start each question in a new booklet
- Write using black or blue pen
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

### Total Marks – 84

- Attempt all questions
- All questions are of equal value

**Question 1** (Start a new booklet)**Marks**

i) Find the value of  $x$  where  $8^{2x-4} = 16^{x-1} \cdot 4^{-x}$  2

ii) Solve for  $x$ :  $\frac{1}{2x-1} \leq 1$ . 2

iii) The sum of the first  $n$  terms of a series is given by  $S_n = \frac{n}{3}(n+1)(n+2)$

a. Show that the  $n$ th term is given by  $T_n = n(n+1)$ . 2

b. Find the sum of the second 50 terms. 1

iv) A geometric series has a limiting sum which has a value of  $\frac{1}{1-w}$

and a common ratio of  $\frac{1}{w}$ . Find the first term. 1

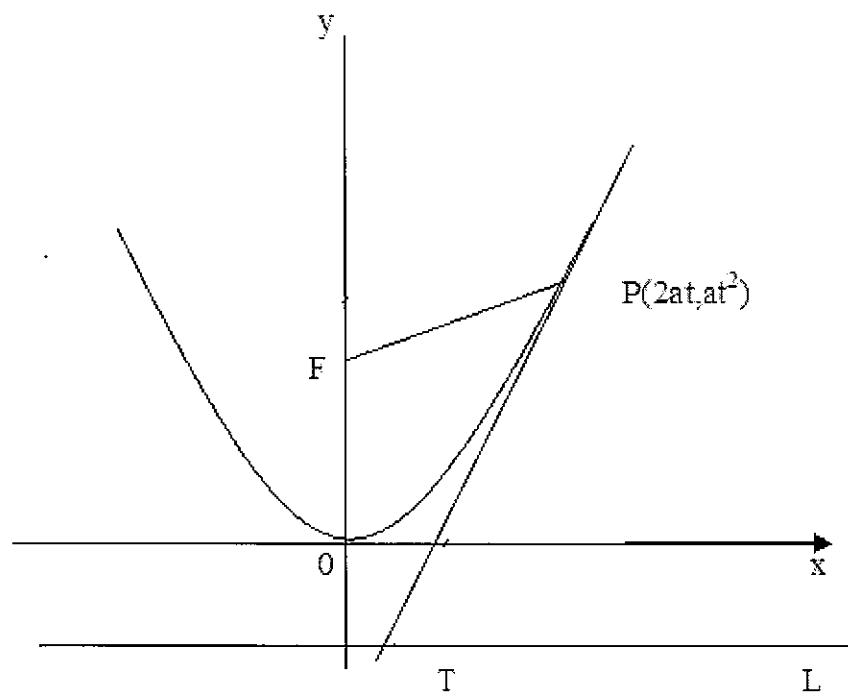
v) Solve for  $x$ , leave your answer in exact form

$$\left(x + \frac{2}{x}\right)^2 - 6\left(x + \frac{2}{x}\right) - 7 = 0$$
 2

vi) Evaluate  $\lim_{x \rightarrow 0} \frac{4 \sin 2x}{3x}$  2

**Question 2** (Start a new booklet)**Marks**

- i) If  $y = (x^2 + 2)^6$ , then find  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  2
- ii) a. Show that  $(\sin A - \cos A)^2 = 1 - \sin 2A$  1
- b. Hence find the exact value of  $(\sin 15^\circ - \cos 15^\circ)$ . 1
- iii) A circle with area  $25 \text{ cm}^2$  has an arc length of 2cm. Find the angle subtended at the centre of the circle by the arc. Answer in exact form. 2
- iv)



The tangent  $tx - y - at^2 = 0$  at the point  $P(2at, at^2)$  on the parabola  $x^2 = 4ay$  cuts the directrix  $L$  at  $T$ .  $F$  is the focus of the parabola.

- a. Find the coordinates of  $T$  1
- b. Show that  $TF$  is perpendicular to  $PF$ . 3
- v) Find the value of  $\lim_{x \rightarrow \infty} \frac{6(10^x) + 9}{4(10^x) - 3}$  2

**Question 3** (Start a new booklet)

**Marks**

i) Evaluate each of the following indefinite integrals

a.  $\int 6xe^{4x^2} dx$  **1**

b.  $\int (2\cos 3x - \sin 2x) dx$  **2**

c.  $\int \frac{1}{3}x(1+2x^2)^3 dx$  **1**

d.  $\int \left(\frac{3x^4}{2x^5 - 3}\right) dx$  **1**

ii) Evaluate

a.  $\int_0^1 \frac{e^x}{2e^x - 1} dx$  **2**

b.  $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x dx$  **2**

iii) Find the area bounded by the curve  $y = 2\sin \frac{\pi}{2}x$ , the  $x$ -axis

and the ordinates  $x = 1$  and  $x = 3$ . **3**

**Question 4** (Start a new booklet)

i) Differentiate with respect to  $x$ :

a.  $\sin(3x^2 - 2)$  **1**

b.  $e^{3x^2 - 8x} + 7x^2$  **1**

c.  $2xe^{6x}$  **2**

d.  $\cot x$  **2**

e.  $\ln\left(\frac{3x-4}{x+5}\right)$  **2**

ii) Six people enter an empty mini bus in which there are 14 seats; 7 on each side. In how many ways can they take their places if;

a. Anyone can occupy any seat. **1**

b. Jacquie sits in one of the four corner seats. **1**

iii) Differentiate  $\sin^2\left(3x - \frac{\pi}{2}\right)$  and find its value where  $x = \frac{\pi}{4}$ . **2**

**Question 5** (Start a new booklet)

**Marks**

i) a. Show that  $\frac{1 - \cos 2x}{1 + \cos 2x} = \tan^2 x$  **2**

b. Hence find the value of  $\tan^2 \left( 22\frac{1}{2} \right)^\circ$ . **2**

ii) a. Express  $\cos x - \sqrt{3} \sin x$  in the form  $A \cos (x + \theta)$  **2**

b. Hence sketch the graph  $y = \cos x - \sqrt{3} \sin x$  for  $0 \leq x \leq 2\pi$  **4**

iii) Solve  $2 \cos^2 \theta + 5 \sin \theta = -1$  where  $0 \leq \theta \leq 2\pi$  **2**

**Question 6** (Start a new booklet)

- i) Use the process of mathematical induction to prove that

$$\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \dots + \frac{1}{x^n} = \frac{1}{x-1} - \frac{1}{x^n(x-1)} \quad \text{for } n \geq 1. \quad 5$$

- ii) a. Prove  $2 \cos^2 x = 1 + \cos 2x$  1

- b. If  $y = 2 \cos^2 2x - 1$ , then evaluate the area between the curve the  
x-axis and the ordinates  $x = \frac{\pi}{16}$  and  $x = \frac{3\pi}{8}$ . 3

- iii) Calculate the volume generated (in simplest exact form), by rotating

about the x-axis the curve  $y = \frac{1}{\sqrt{4-x}}$  between  $x = 0$  and  $x = 3\frac{3}{4}$ . 3

**Question 7** (Start a new booklet)

**Marks**

- i) A polynomial  $P(x)$  has the following features.  $P(x)$  is monic and of degree 2.  $P(3) = P(-5)$ .  $P(x)$  has a remainder  $-6$  when divided by  $x$ .

Find  $P(x)$ .

**3**

- ii) By using the substitution  $u = x^2$ , evaluate  $\int x \cos x^2 \, dx$

**3**

- iii) Seven adults (including Joan and Mary) are to be selected at random from a group of ten adults.

a. How many unordered selections are possible if no restrictions exist?

**1**

b. How many unordered selections are possible if Joan and Mary are not in the same group?

**2**

- iv) The area between the curve  $y = \frac{3}{x+4}$ , the y-axis and the ordinates  $x = -3\frac{3}{4}$  and  $x = -3\frac{1}{2}$  is rotated about the y-axis. Find the volume generated in exact form.

**3**

**End of Paper**





# Question 1

$$i) 8^{2x-4} = 16^{x-1} \cdot 4^{-x}$$

$$2^{6x-12} = 2^{2x-4} \quad \checkmark$$

$$6x-12 = 2x-4$$

$$x = 2 \quad \checkmark$$

$$ii) \frac{1}{2x-1} \leq 1 \quad (x \neq \frac{1}{2})$$

$$\left(\frac{1}{2x-1}\right)(2x-1)^2 \leq 1 \times (2x-1)^2$$

$$4x^2 - 4x + 1 \geq 2x - 1$$

$$2x^2 - 3x + 1 \geq 0 \quad \checkmark$$

$$(2x-1)(x-1) \geq 0$$

$$x < \frac{1}{2} \text{ and } x \geq 1 \quad \checkmark$$

$$iii) a) S_n = \frac{n}{3}(n+1)(n+2)$$

$$S_1 = 2, S_2 = 8, S_3 = 20, S_4 = 40$$

$\therefore$  Series is

$$2, 6, 12, 20 \quad \checkmark$$

$$1 \times 2, 2 \times 3, 3 \times 4, 4 \times 5, \dots$$

$$T_1 \quad T_2 \quad T_3 \quad T_4$$

$$\therefore T_n = n(n+1) \quad \checkmark$$

$$b) \text{Sum 2nd 50} = S_{100} - S_{50}$$

$$= \left(\frac{100}{3}\right)(101)(102) - \left(\frac{50}{3}\right)(51)(52)$$

$$\checkmark = 299200$$

IV

$$S_{\infty} = \frac{a}{1-r}$$

$$\frac{1}{1-\omega} = \frac{a}{1-\frac{1}{\omega}}$$

$$\frac{1-\frac{1}{\omega}}{1-\omega} = a$$

$$a = \frac{\omega-1}{-\omega^2+\omega}$$

$$= -\frac{1}{\omega} \quad \checkmark$$

$$V \text{ let } A = x + \frac{2}{x}$$

$$A^2 - 6A - 7 = 0$$

$$A = 7 \text{ or } A = -1$$

$$x + \frac{2}{x} = 7 \text{ or } x + \frac{2}{x} = -1 \quad \checkmark$$

$$x = \frac{7 \pm \sqrt{49}}{2} \quad \checkmark$$

$$VI) \lim_{x \rightarrow 0} \frac{4 \sin 2x}{3x}$$

$$= \frac{4}{3} \times 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \quad \checkmark$$

$$= \frac{8}{3} \times 1$$

$$= \frac{8}{3} \quad \checkmark$$

## Question 2

$$i) \quad y = (x^2 + 2)^6$$

$$y' = 12x(x^2 + 2)^5 \quad \checkmark$$

$$y'' = 12(x^2 + 2)^5 + 120x^2(x^2 + 2)^4 \quad \checkmark$$

ii)  $q$

$$(\sin A - \cos A)^2$$

$$= \sin^2 A + \cos^2 A - 2 \sin A \cos A \quad \checkmark$$

$$= 1 - \sin 2A$$

$$b \quad (\sin 15 - \cos 15)$$

$$= \sqrt{1 - \sin(2 \times 15)}$$

$$= \sqrt{1 - \sin 30}$$

$$= \frac{1}{\sqrt{2}} \quad \checkmark$$

$$iii) \quad \pi r^2 = 25$$

$$r = \frac{5}{\sqrt{\pi}} \quad \checkmark$$

$$L = 10$$

$$2 = \frac{5}{\sqrt{\pi}} \quad \checkmark$$

$$\theta = \frac{2\sqrt{\pi}}{5} c$$

$\therefore$  The angle at the centre is  $\frac{2\sqrt{\pi}}{5} c \quad \checkmark$

$$iv) \quad q \quad tx - y - at^2 = 0 \quad \text{--- (1)}$$

$$y = -a \quad \text{--- (2) The directrix}$$

sub (2) into (1)

$$tx = at^2 - a$$

$$x = \frac{at^2 - a}{t} \quad \checkmark$$

$$\therefore T = \left( \frac{at^2 - a}{t}, -a \right) \quad \checkmark$$

$$iv) \quad b. \quad F = (0, a)$$

$$M_{PF} = \frac{at^2 - a}{2at} = \frac{t^2 - 1}{2t} \quad \checkmark$$

$$M_{TF} = \frac{-a - a}{\frac{at^2 - a}{t}}$$

$$= \frac{-2at}{a(t^2 - 1)}$$

$$= \frac{-2t}{t^2 - 1} \quad \checkmark$$

$$M_{PF} \times M_{TF} = \frac{t^2 - 1}{2t} \times \frac{-2t}{t^2 - 1}$$

$$= -1.$$

Hence  $TF \perp PF \quad \checkmark$

$$v) \quad \lim_{n \rightarrow \infty} \frac{6(10^n) + 9}{4(10^n) - 3}$$

$$= \lim_{n \rightarrow \infty} \frac{6 + \frac{9}{10^n}}{4 - \frac{3}{10^n}} \quad \checkmark$$

$$= \frac{6 + 0}{4 - 0}$$

$$= 1\frac{1}{2} \quad \checkmark$$

### Question 3

i) a  $\int 6x e^{4x^2} dx$   
 $= \frac{3}{4} e^{4x^2} + C \quad \checkmark$

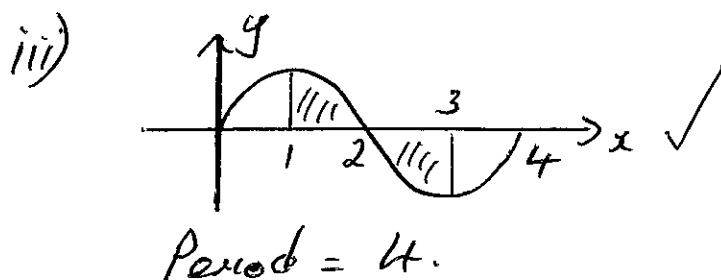
b  $\int (2 \cos 3x - \sin 2x) dx$   
 $= \frac{2}{3} \sin 3x + \frac{1}{2} \cos 2x + C \quad \checkmark$

c  $\frac{1}{3} \int x(1+2x^2)^3 dx$   
 $= \frac{1}{48} (1+2x^2)^4 + C \quad \checkmark$

d  $\int \frac{3x^4}{2x^5-3} dx$   
 $= \frac{3}{10} \ln(2x^5-3) + C \quad \checkmark$

ii) a  $\int_0^1 \frac{e^x}{2e^x-1} dx$   
 $= \frac{1}{2} \int_0^1 \frac{2e^x}{2e^x-1} dx$   
 $= \frac{1}{2} [\ln(2e^x-1)]_0^1 \quad \checkmark$   
 $= \frac{1}{2} [\ln(2e-1)] \quad \checkmark$

ii) b  $\int_0^{\pi/2} \sin^2 x \cos x dx$   
 $= \left[ \frac{1}{3} \sin^3 x \right]_0^{\pi/2} \quad \checkmark$   
 $= \frac{1}{3} \left[ \sin^3 \frac{\pi}{2} - \sin^3 0 \right]$   
 $= \frac{1}{3} \quad \checkmark$



~~$A = \int_1^3 2 \sin \frac{\pi}{2} x dx$~~   
 ~~$= \left[ -\frac{2}{\pi} \cos \frac{\pi}{2} x \right]_1^3$~~   
 ~~$=$~~

$A = 2 \int_1^3 2 \sin \frac{\pi}{2} x dx \quad \checkmark$   
 $= 4 \left[ -\cos \frac{\pi}{2} x \right]_1^3$   
 $= 4 \left[ -\frac{2}{\pi} (\cos \frac{3\pi}{2} - \cos \frac{\pi}{2}) \right]$   
 $= \frac{8}{\pi} \text{ units}^2 \quad \checkmark$

## Question 4

i) a  $\frac{d}{dx} \sin(3x^2-2)$   
 $= 6x \cos(3x^2-2) \checkmark$

b  $\frac{d}{dx} (e^{3x^2-8x} + 7x^2)$   
 $= (6x-8)(e^{3x^2-8x}) + 14x \checkmark$

c  $\frac{d}{dx} (2x e^{6x})$   
 $= (2x)(6e^{6x}) + (e^{6x})(2) \checkmark$   
 $= 2e^{6x}(6x+1) \checkmark$

d  $\frac{d}{dx} (\cot x)$   
 $= \frac{d}{dx} \left( \frac{\cos x}{\sin x} \right)$   
 $= \frac{\sin x(-\sin x) - (\cos x)(\cos x)}{\sin^2 x} \checkmark$   
 $= \frac{-1}{\sin^2 x} \checkmark$

e  $\frac{d}{dx} \left[ \ln \left( \frac{3x-4}{x+5} \right) \right]$   
 $= \frac{d}{dx} [\ln(3x-4) - \ln(x+5)] \checkmark$   
 $= \frac{3}{3x-4} - \frac{1}{x+5} \checkmark$

ii)

a  ${}^{14}P_6 = 2,162,160 \checkmark$

b  $(13 \times 12 \times 11 \times 10 \times 9) \times 4 \checkmark$   
 $= 617,760$

iii) let  $y = \sin^2 \left( 3x - \frac{\pi}{2} \right)$   
 $y' = 2 \sin \left( 3x - \frac{\pi}{2} \right)$   
 $\times \cos \left( 3x - \frac{\pi}{2} \right) \times 3$   
 $\checkmark = 6 \sin \left( 3x - \frac{\pi}{2} \right) \cos \left( 3x - \frac{\pi}{2} \right)$

when  $x = \frac{\pi}{4}$ ,

$y' = 6 \sin \frac{\pi}{4} \cos \frac{\pi}{4}$   
 $= 6 \times \frac{1}{2}$   
 $\checkmark = 3$

## Question 5

i) a) LHS =  $\frac{1 - \cos 2x}{1 + \cos 2x}$

$$= \frac{1 - (1 - 2\sin^2 x)}{1 + (2\cos^2 x - 1)} \checkmark$$

$$= \frac{2\sin^2 x}{2\cos^2 x}$$

$$= \tan^2 x \checkmark$$

$$= \text{RHS}$$

b)  $\tan^2 22\frac{1}{2} = \frac{1 - \cos 45}{1 + \cos 45} \checkmark$

$$= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}$$

$$= \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \checkmark$$

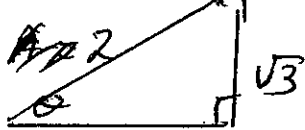
ii)  $A \cos(x + \theta)$

$$= A \cos x \cos \theta - A \sin x \sin \theta$$

$$= 1 \cos x - \sqrt{3} \sin x$$

$\therefore A \cos \theta = 1 \checkmark, A \sin \theta = \sqrt{3}$

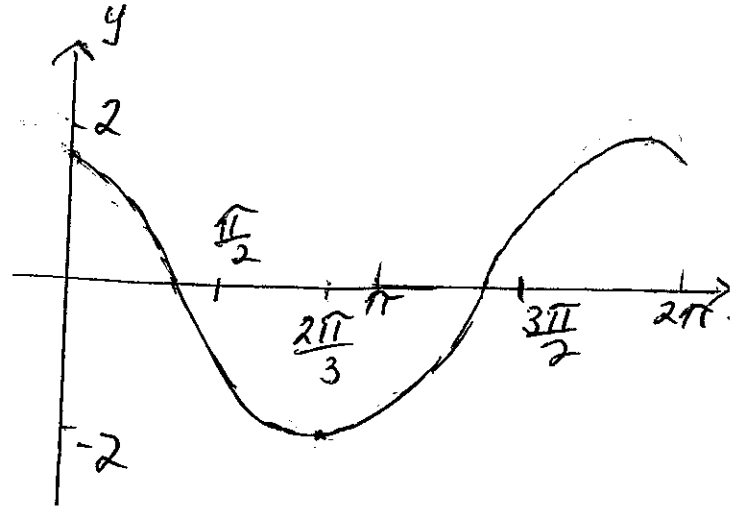
$\therefore \tan \theta = \sqrt{3}$



$\therefore A = 2, \theta = \frac{\pi}{3}$

$\therefore \cos x - \sqrt{3} \sin x = 2 \cos(x + \frac{\pi}{3}) \checkmark$

ii) b



$y = \cos x - \sqrt{3} \sin x$

$$= 2 \cos(x + \frac{\pi}{3})$$

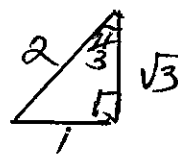
Amplitude = 2 }  $\checkmark$   
 Period =  $2\pi$  }  $\checkmark$   
 labelling }  $\checkmark$   
 shape }  $\checkmark$   
 accuracy }  $\checkmark$

iii)  $2 \cos^2 \theta + 5 \sin \theta = -1$

$2 \sin^2 \theta - 5 \sin \theta - 3 = 0$

$(2 \sin \theta + 1)(\sin \theta - 3) = 0 \checkmark$

$2 \sin \theta + 1 = 0, \sin \theta - 3 = 0$   
 $\sin \theta = -\frac{1}{2}$  No solution



|               |                 |
|---------------|-----------------|
| S             | A               |
| $\frac{1}{2}$ | $\frac{\pi}{6}$ |

$\theta = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$   
 $= \frac{7\pi}{6}, \frac{11\pi}{6} \checkmark$

## Question 6.

i. For  $n=1$

$$\text{LHS} = \frac{1}{x}, \quad \text{RHS} = \frac{1}{x-1} - \frac{1}{x(x-1)} = \frac{x-1}{x(x-1)} = \frac{1}{x} = \text{LHS} \quad \checkmark$$

$\therefore$  true for  $n=1$ .

Assume true for  $n=k$ .

$$\text{i.e. } \frac{1}{x} + \frac{1}{x^2} + \dots + \frac{1}{x^k} = \frac{1}{x-1} - \frac{1}{x^k(x-1)} \quad \text{--- (1) } \checkmark$$

Required to show true for  $n=k+1$

$$\text{i.e. } \frac{1}{x} + \frac{1}{x^2} + \dots + \frac{1}{x^k} + \frac{1}{x^{k+1}} = \frac{1}{x-1} - \frac{1}{x^{k+1}(x-1)}$$

For  $n=k+1$

$$\text{LHS} = \frac{1}{x} + \frac{1}{x^2} + \dots + \frac{1}{x^k} + \frac{1}{x^{k+1}} \quad \checkmark$$

$$= \frac{1}{x-1} - \frac{1}{x^k(x-1)} + \frac{1}{x^{k+1}} \quad \text{from (1)}$$

$$= \frac{1}{x-1} - \frac{1}{x^k(x-1)} + \frac{1}{x^k \cdot x}$$

$$= \frac{1}{x-1} + \frac{-x + x-1}{x^k \cdot x \cdot (x-1)}$$

$$= \frac{1}{x-1} - \frac{1}{x^{k+1}(x-1)} \quad \checkmark$$

$$= \text{RHS.}$$

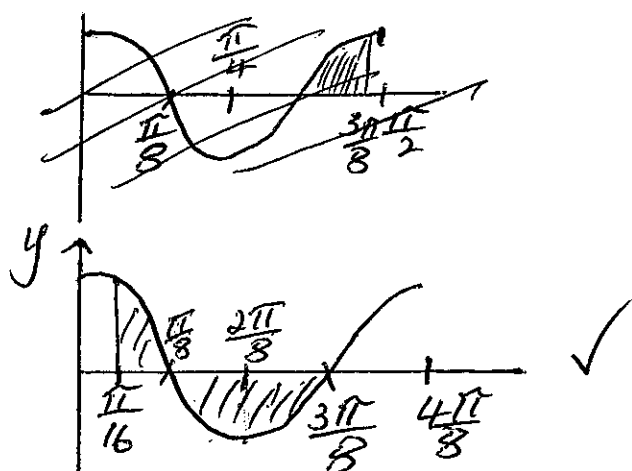
$\therefore$  If true for  $n=k$  then it is true for  $n=k+1$ .  
true for  $n=1$ ,  $\therefore$  true for  $n=1+1=2$   
true for  $n=2$ ,  $\therefore$  true for  $n=2+1=3$   $\checkmark$   
and so on.

$\therefore$  true for all  $n \geq 1$ .

## Question 6 (continued)

ii) a  $\text{RHS} = 1 + \cos 2x$   
 $= 1 + 2\cos^2 x - 1$   
 $= 2\cos^2 x$   
 $= \text{LHS} \quad \checkmark$

b  $y = 2\cos^2 x - 1$   
 $y = \cos 4x$   
 Period  $= \frac{\pi}{2}$ .



$$A = \int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \cos 4x \, dx + \left| \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \cos 4x \, dx \right| \checkmark$$

$$= \left[ \frac{1}{4} \sin 4x \right]_{\frac{\pi}{16}}^{\frac{\pi}{8}} + \left| \left[ \frac{1}{4} \sin 4x \right]_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \right|$$

$$= \left( \frac{1}{4} - \frac{\sqrt{2}}{8} \right) + \left| \left( -\frac{1}{4} - \frac{1}{4} \right) \right|$$

$$= \left( \frac{3}{4} - \frac{\sqrt{2}}{8} \right) \text{ units}^2 \quad \checkmark$$

iii)  $y = \frac{1}{\sqrt{4-x}}$   
 $y^2 = \frac{1}{4-x} \quad \checkmark$

$$V = \int_0^{\frac{15}{4}} \pi y^2 \, dx$$

$$= -\pi \int_0^{\frac{15}{4}} \frac{-1}{4-x} \, dx \quad \checkmark$$

$$= -\pi \left[ \ln(4-x) \right]_0^{\frac{15}{4}}$$

$$= -\pi \left[ \ln \frac{1}{4} - \ln 4 \right]$$

$$= 4\pi \ln 2 \text{ units}^3 \quad \checkmark$$



## Question 7

$$i) P(x) = ax^2 + bx + c \quad a = 1$$

$$= x^2 + bx + c$$

$$P(3) = 9 + 3b + c$$

$$P(-5) = 25 - 5b + c$$

$$P(3) = P(-5)$$

$$\therefore 9 + 3b + c = 25 - 5b + c$$

$$b = 2$$

$$\therefore P(x) = x^2 + 2x + c$$

$$P(0) = -6$$

$$\therefore c = -6$$

$$\therefore P(x) = x^2 + 2x - 6$$

$$ii) u = x^2 \therefore du = 2x dx$$

$$\therefore \int x \cos x^2 dx$$

$$= \frac{1}{2} \int 2x \cos x^2 dx$$

$$= \frac{1}{2} \int \cos u du$$

$$= \frac{1}{2} (-\sin u) + c$$

$$= -\frac{1}{2} \sin x^2 + c$$

iii)

a.

$${}^{10}C_7 = 120$$

b. There are 3 possibilities

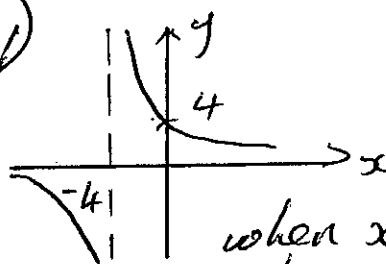
✓ Joan in Mary not  
 Mary in Joan not  
 Joan and Mary not in.

$$\text{Total Number of Selections}$$

$$= {}^8C_6 + {}^8C_6 + {}^8C_7$$

$$= 64$$

iv)



$$\text{when } x = -3\frac{3}{4}, y = 12$$

$$\text{when } x = -3\frac{1}{2}, y = 6$$

$$x + 4 = \frac{3}{y}$$

$$x^2 = \left(\frac{9}{y^2} - \frac{24}{y} + 16\right)$$

$$V = \pi \int x^2 dy$$

$$= \pi \int_6^{12} \left(\frac{9}{y^2} - \frac{24}{y} + 16\right) dy$$

$$= \pi \left[ \frac{9}{-y} - 24 \ln y + 16y \right]_6^{12}$$

$$= \left[ 9 \cdot \frac{3}{4} - 24 \ln 12 + 24 \ln 6 \right] \pi$$

units<sup>3</sup>